

Task 2 Report -- AI Study Assistant

Artificial Intelligence Assignment #1

Build a Simple AI Agent Web App

Student Information

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Semester: 5th Semester
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1. Introduction

This project is a simple AI-powered Study Assistant developed using Python, Flask, HTML, CSS, and the Groq API. The application allows users to ask study-related questions and receive AI-generated responses in simple beginner-friendly language.

The purpose of this project is to demonstrate how Large Language Models (LLMs) can be integrated into a web application using a free API platform.

The AI assistant acts as a virtual tutor that helps users understand educational concepts related to Artificial Intelligence, Machine Learning, Python programming, and other academic topics.

2. Objective

The objective of this project was to:

- Build a simple AI-powered web application
- Integrate a free LLM API
- Create a user-friendly interface using Flask
- Implement prompt engineering using a system prompt
- Allow users to interact with the AI assistant through a web browser

3. Technologies Used

Technology	Purpose
Python	Backend programming
Flask	Web framework
Groq API	AI response generation

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HTML	Frontend structure
CSS	Styling and UI
python-dotenv	Environment variable handling

4. API Platform Used

Groq API

The Groq API was used to power the AI Study Assistant. Groq provides fast and free access to Large Language Models.

Model Used: llama-3.1-8b-instant

5. Project Features

The AI Study Assistant includes the following features:

- User-friendly web interface
- AI-generated educational responses
- Beginner-friendly explanations
- Integration with Groq LLM API
- Secure API key handling using .env
- Real-time interaction between user and AI

6. System Prompt

The following system prompt was used to define the AI assistant's behavior:

You are a friendly AI Study Assistant.

Explain concepts in simple beginner-friendly language.

This prompt helps the AI provide educational and easy-to-understand responses.

7. Project Structure

```
task2-ai-agent/  
|  
|-- app.py  
|-- requirements.txt
```

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```
| -- .env
| -- .env.example
| -- README.md
|
| -- templates/
|   |-- index.html
|
| -- static/
|   |-- style.css
```

8. Working Procedure

Step 1 -- User Input

The user enters a study-related question into the text box on the web page.

Step 2 -- Flask Backend

The Flask application receives the user input using a POST request.

Step 3 -- API Request

The application sends the question to the Groq API along with the system prompt.

Step 4 -- AI Response

The Groq model generates a response based on the user's query.

Step 5 -- Display Output

The response is displayed on the webpage for the user.

9. Sample Queries Tested

The application successfully handled the following queries:

1. What is Artificial Intelligence?
2. Explain Machine Learning
3. Difference between AI and ML
4. What is Python?
5. Explain supervised learning

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10. Challenges Faced

Problem	Solution
Python not detected	Installed Python and added to PATH
Flask module error	Installed required packages
API key issue	Configured .env correctly
405 Method Error	Fixed Flask route methods
Deprecated model error	Updated Groq model name

11. Results

The AI Study Assistant successfully generated accurate and beginner-friendly responses for educational questions. The web interface functioned properly, and the API integration worked successfully.

The project achieved the objective of building an AI-powered web application using a free LLM API.

12. Conclusion

This project provided practical experience in integrating Artificial Intelligence with web development. It helped in understanding APIs, Flask routing, environment variables, frontend-backend communication, and prompt engineering.

The AI Study Assistant can be further improved by adding advanced features such as voice input, chat history, dark mode, and file upload support.

13. Future Improvements

Possible future improvements include:

- Chat-style UI
- Voice interaction
- Multi-language support
- User authentication
- AI conversation history
- Better UI/UX design

14. References

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1. Flask Documentation -- <https://flask.palletsprojects.com/>
2. Groq API Documentation -- <https://console.groq.com/docs>
3. Python Official Website -- <https://www.python.org/>